

Lecture 7

Last: $a|b$, $a = dq + r$, $a \equiv b \pmod{m} \iff$
 $0 \leq r < d$ (WOP) $a \bmod m = b \bmod m \iff$
 $m|b-a$. Prop 2

I Modular

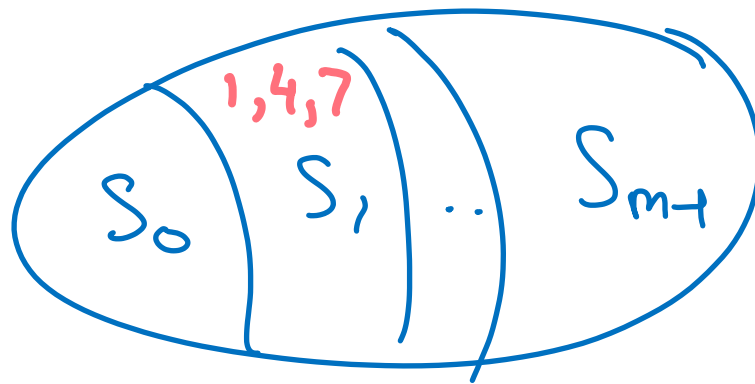
II Primes

III GCD

Familiar: (Prop 3, lec 6): Can substitute, add, mult congruences.

Unfamiliar: $1 \equiv 4 \equiv 7 \equiv 10 \equiv 13 \equiv \dots \pmod{3}$
 $m=3$

Cartoon:



$\mathbb{Z} = \bigcup_{j=0}^{m-1} S_j$

$S_j = \{a \in \mathbb{Z} : a \equiv j \pmod{m}\}$

Poll: Is $\forall x > 1, a \equiv b \pmod{m}$ then $x^a \equiv x^b \pmod{m}$?

False: - what if $a, b < 0$? $x^{-1} \equiv x^{-4} \pmod{m}$?

$\exists x, a, b$ - $x=2, m=3$ concrete.
($a \equiv b \pmod{m}$) $2^a \equiv 2^b \pmod{3} \iff 3 \mid 2^b - 2^a$.
 $\wedge x^a \not\equiv x^b \pmod{m}$) Choose $b=4, a=1$:
 $\Downarrow$
 $3 \mid 16 - 2$?
 $3 \nmid 14$ False.

When in doubt, read the definitions.

Poll: $\forall m, a, b, c, d$ ($a \equiv b \pmod{m}$ $\wedge$ $b \equiv c+d \pmod{m}$)
 $\longrightarrow a^2 \equiv c^2 + d^2 \pmod{m}$!

$$a \equiv b \equiv c+d \pmod{m}.$$

$$a^2 \equiv (c+d)^2 \equiv c^2 + 2cd + d^2 \pmod{m}.$$
$$\stackrel{?}{\equiv} c^2 + d^2 \pmod{m}.$$

If $m=2$: $c^2 + 2cd + d^2 \equiv c^2 + d^2 \pmod{2}.$
 $2 \equiv 0 \pmod{2}$

If $m=3$, $c=d=1$: $c^2 + 2cd + d^2 \equiv 1 + 2 + 1$
 $\equiv 4$
 $\equiv 1 \pmod{3}$

but $c^2 + d^2 \equiv 2 \pmod{3}.$

Congruences depend on m .

Representations of Numbers:

ex: $596257 = 7 \cdot 10^0 + 5 \cdot 10^1 + 2 \cdot 10^2 + 6 \cdot 10^3 + 9 \cdot 10^4 + 5 \cdot 10^5$

Want to know $596257 \bmod 3$. ① Division.

② Observe $10 \equiv 1 \pmod{3}$.
 $\forall p > 1 \quad 10^p \equiv 1^p \equiv 1 \pmod{3}$.

$$\begin{aligned} 7 \cdot 10^0 + 5 \cdot 10^1 + 2 \cdot 10^2 + 6 \cdot 10^3 + 9 \cdot 10^4 + 5 \cdot 10^5 &\stackrel{\text{Substitute Cong.}}{\equiv} 7 \cdot 1 + 5 \cdot 1 + 2 \cdot 1 + 6 \cdot 1 + 9 \cdot 1 + 5 \cdot 1 \pmod{3} \\ &\equiv 1 + 2 + 2 + 0 + 0 + 2 \\ &\equiv 1 \pmod{3}. \end{aligned}$$

Much faster than division!

Prop: A decimal number is divisible by 3
 $\longleftrightarrow$ the sum of its digits is divisible by 3.

(Q) What about other numbers?

(A) This is particularly simple.

Hw: Algorithm for finding binary expansion of an integer (powers of 2). [Algorithm 1].

II Primes — Quiz extended til 8pm tomorrow.

Defn: An integer $n > 1$ is prime if it has no positive divisors other than 1 and itself.

A number that is not prime is called composite
($\exists a, b > 1 : n = ab$)

Useful prop.

Lemma 1: If $c > 1$ is composite, there is a prime p such that $p | c$.

Proof: Assume $c > 1$ is composite.

Let $S = \{ a \in \mathbb{Z}_+ : a | c \text{ and } a > 1 \} \subseteq \mathbb{Z}_+$.

By defn, $S \neq \emptyset$, so by the WOP we can choose a least element $p \in S$.

Assume for contradiction that p is composite.

By defn, we can choose $b > 1$ such that $b | p$, $b < p$.

By Leib Prop 1, we conclude $b | c$ so $b \in S$.

This contradicts the minimality of p . $\square$

$p | c$
 $b | p$

Theorem (Euclid, 300 BC): There are infinitely many primes.

Proof: Assume for contradiction there are finitely many primes $p_1, p_2, p_3, \dots, p_n$. } all primes.

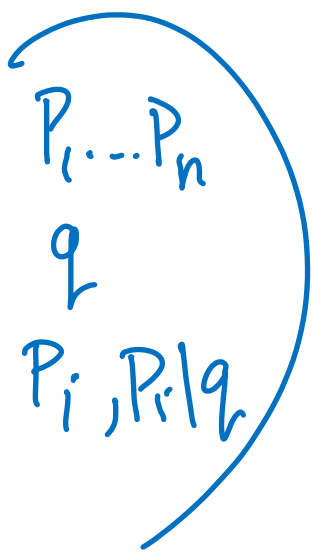
Consider $q = p_1 p_2 p_3 \dots p_n + 1$.

Observe that q is not prime since $q > p_i$ for all $i = 1 \dots n$.

Therefore q is composite and by Lemma 1, we can choose a p_i such that $p_i | q$.

By construction, we also have $p_i | p_1 p_2 \dots p_n$, i.e.

$p_i | (q - 1)$. By Lec 6, $p_i | q - (q - 1)$ i.e. $p_i | 1$, which is absurd. $\square$



II GCD.

Defn: If $a, b \in \mathbb{Z}$, not both zero, the greatest common divisor of a and b , denoted $\boxed{\gcd(a, b)}$ is the largest $d \in \mathbb{Z}_+$ s.t. $d|a$ and $d|b$.

ex: $a = 10$.
 $b = 36$.
 $\gcd(a, b) = 2$.

divisors of $a = \{1, 2, 5, 10\}$

" $b = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$

Surprise: Computing sets of divisors is hard but
computing $\gcd(a, b)$ is easy.

Euclidean Algorithm:

Key Observation : $\text{If } a > b > 1 \text{ and } a = bq + r$
 $0 \leq r < b$
then $\gcd(a, b) = \gcd(b, r).$

ex2: $a = 252, b = 198$

$\gcd(252, 198) = \text{key.}$

$\gcd(198, 54) =$

$\gcd(54, 36) =$

$\gcd(36, 18) =$

18

Division

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1.36 + 18$$

$$z_6 = 2.1840$$

next time.

Algorithm: Apply the key observation repeatedly until $r=0$.

(Q) Why does this always terminate?

(A) The second number is decreasing ($r < b$) and stays nonnegative ($r \geq 0$).

The key observation follows from.

Lemma 2: if $a > b > 1$ and $a = bq + r$, $0 \leq r < b$ then

"set of common divisors is preserved". $\forall d \in \mathbb{Z}_+ (d|a \text{ and } d|b \iff d|b \text{ and } d|r)$.

Proof: ($\rightarrow$) Assume $a > b > 1$, $a = bq + r$, $0 \leq r < b$.

Assume $d \in \mathbb{Z}_+$, $d|a$, and $d|b$.

Observe that
so $d|a$ and $d|bq$
 $d|a - bq$ i.e. $d|r$.

$$\begin{array}{l} d|a, d|b \\ a = bq + r \end{array}$$

Thus, $d|b$ and $d|r$, as desired.

($\leftarrow$) $\langle \text{HW} \rangle$

□

Bézout's Theorem: If $a, b \in \mathbb{Z}$ not both zero there exist $s, t \in \mathbb{Z}$ such that $\gcd(a, b) = sa + tb$.
[Strong Bézout: $\gcd(a, b) = \min(S)$] "linear combination"

Proof: Assume $a, b \in \mathbb{Z}$ not both zero.

Consider $S = \{sa + tb : s, t \in \mathbb{Z}, sa + tb \geq 1\} \subseteq \mathbb{Z}_+$.

Observe that $S \neq \emptyset$ since $a^2 + b^2 \in S$. $\leftarrow \begin{matrix} s=a \\ t=b \end{matrix}$

By the wop, we can choose a least $g \in S$
s.t. $g = sa + tb$.

We will show that $g = \gcd(a, b)$.

$\begin{matrix} a, b \\ S \\ g = sa + tb \\ a = gq + r. \end{matrix}$

Claim 1: $g|a$ and $g|b$: Assume for contradiction $g \nmid a$.

Choose r such that $a = gq + r$ and $0 < r < g$.

Observe that

$$\begin{aligned} r &= a - gq \\ &= a - (sa + tb)q \\ &= (1 - qs)a + (-tq)b \end{aligned}$$

and $r > 0$, so $r \in S$, contradicting the minimality of g .

The proof of $g|b$ is completely analogous.

Claim 2: If $d \in \mathbb{Z}_+$, $d|a$, and $d|b$ then $d|g$.

<HW>.

$\therefore g = \gcd(a, b)$ as desired.



How to compute s, t ?

